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Rule-Based Conversational Chatbot in Python

A simple Python chatbot that takes user input, understands common messages, responds using predefined rules, and continues the conversation until the user exits.

1. Project Overview

The goal of this project is to create a simple conversational chatbot using Python. The chatbot communicates with the user through the terminal. It first asks for the user's name and then waits for messages. Based on the message, it gives a suitable response. If it does not recognize the message, it gives a helpful fallback response. The conversation continues until the user types **bye** or **exit**.

2. How the Chatbot Works

The chatbot follows a simple process. First, it gets the user's name. During the conversation, it receives a message using `input()`. The message is cleaned using `lower().strip()`, so inputs such as **HELLO** and **hello** can be handled more easily. The chatbot then checks different conditions using **if**, **elif**, and **else**. The **in** operator makes the chatbot more flexible, so messages such as **hello bot** can also be recognized as greetings.

3. Python Code

```
name = input("What is your name? ").strip()

print("Hello", name, "! I am a simple chatbot.")
print("You can ask me about Python, study, or my name.")
print("Type 'bye' or 'exit' to end the conversation.")

while True:
    message = input("You: ").lower().strip()

    if "hello" in message or "hi" in message:
        print("Bot: Hello", name, "! How can I help you?")

    elif "python" in message:
        print("Bot: Python is a high-level programming language "
              "used to build many types of applications.")

    elif "study" in message:
        print("Bot: Make a simple study plan, learn one topic at a time, "
              "and practice regularly.")

    elif "your name" in message:
        print("Bot: I am a simple rule-based Python chatbot.")

    elif "how are you" in message:
        print("Bot: I am doing well. Thanks for asking!")

    elif "bye" in message or "exit" in message:
        print("Bot: Goodbye", name, "! Have a great day.")
        break

    else:
        print("Bot: Sorry, I do not understand that message. "
              "Please try another question.")
```

4. Requirement Checklist

No.	Requirement	How it is satisfied
1	Ask the user's name	The chatbot asks for the name before starting the main conversation.
2	Use while True	while True keeps the conversation running continuously.
3	Use input()	input() receives the user's messages.
4	Clean the input	.lower().strip() converts the message to lowercase and removes extra spaces.
5	Recognize greetings	The chatbot recognizes both hello and hi.
6	Answer 3+ topics	It responds to Python, study, name, and how are you.
7	Use if / elif / else	These statements decide which response should be given.
8	Use break	break ends the loop when the user types bye or exit.
9	Use user's name	The chatbot uses the user's name in greeting and goodbye messages.
10	Unknown message	The else block gives a useful response when the message is not recognized.
Challenge	Use in operator	The chatbot can recognize variations such as hello bot or hey hello.

5. Sample Test Run

What is your name? Siva
Hello Siva ! I am a simple chatbot.

You: hello
Bot: Hello Siva ! How can I help you?

You: HELLO
Bot: Hello Siva ! How can I help you?

You: hello bot
Bot: Hello Siva ! How can I help you?

You: hi
Bot: Hello Siva ! How can I help you?

You: What is Python?
Bot: Python is a high-level programming language used to build many types of applications.

You: I want to study
Bot: Make a simple study plan, learn one topic at a time, and practice regularly.

You: I like cricket
Bot: Sorry, I do not understand that message. Please try another question.

You: bye
Bot: Goodbye Siva ! Have a great day.

6. Target Flow

INPUT → PROCESS → RESPOND → REPEAT → EXIT

The user gives a message as input. The chatbot processes and cleans it, checks the predefined rules, gives an appropriate response, and repeats the same process. When the user enters **bye** or **exit**, the chatbot uses **break** and ends the conversation.

7. Conclusion

This project demonstrates the basic idea behind a rule-based conversational chatbot. It uses simple Python concepts such as input handling, strings, conditional statements, loops, the in operator, and break. Although the chatbot uses predefined rules instead of artificial intelligence, it can still respond naturally to several common messages and can be extended with more topics and responses.